

Name \_\_\_\_\_ Teacher \_\_\_\_\_



## **GOSFORD HIGH SCHOOL**

**2012**

### **HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 2 MATHEMATICS – EXTENSION 1**

**Duration-** 90 minutes plus 5 minutes reading time

|                                                                                                                                                                                                                              |            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Question 1 (Multiple choice)</b> 6 questions worth 1 mark each.<br>(Answer this section on the <b>test paper</b> .<br><b>Circle the correct answer</b> )                                                                  | <b>/6</b>  |
| <b>Questions 2 to 7 (free response questions)</b> These must be answered on your <b>own paper</b> . Start a new page for <b>each</b> question. Each question must be stapled separately Each question worth <b>9 marks</b> . |            |
| <b>Question 2 Preliminary topics</b>                                                                                                                                                                                         | <b>/9</b>  |
| <b>Question 3 Iterative methods/polynomials</b>                                                                                                                                                                              | <b>/9</b>  |
| <b>Question 4 Mathematical induction</b>                                                                                                                                                                                     | <b>/9</b>  |
| <b>Question 5 Parametric representation</b>                                                                                                                                                                                  | <b>/9</b>  |
| <b>Question 6 Methods of integration</b>                                                                                                                                                                                     | <b>/9</b>  |
| <b>Question 7 Circle Geometry</b>                                                                                                                                                                                            | <b>/9</b>  |
| <b>TOTAL</b>                                                                                                                                                                                                                 | <b>/60</b> |

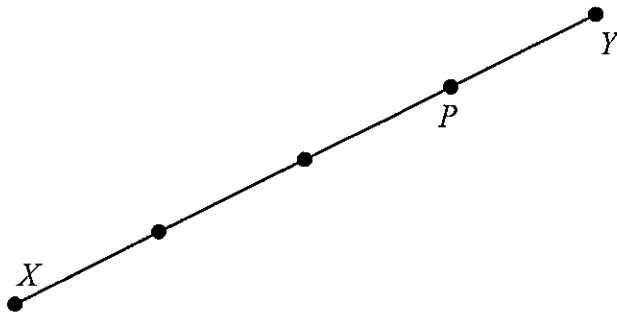
## QUESTION 1

6 marks

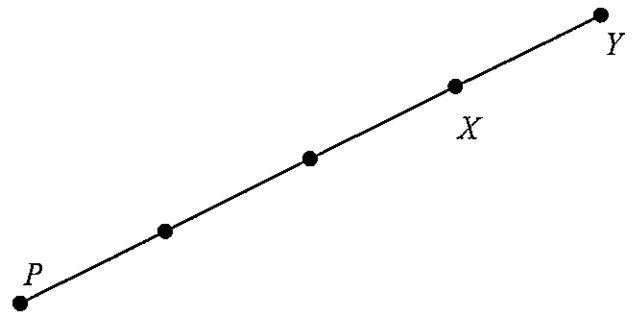
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- i) Which one of the following diagrams best represents a point  $P$  dividing an interval  $XY$  externally in the ratio  $3 : 1$

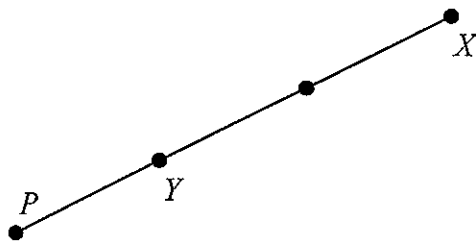
A)



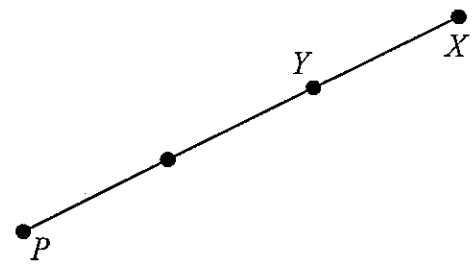
B)



C)



D)



- ii) The curve  $y = f(x)$  is defined for all values of  $x$ .  
Given that  $a < b < c$  and  $K$  is a constant, which one of the following is NOT necessarily always true.

A)  $\int_b^a f(x)dx = -\int_a^b f(x)dx$

B)  $\int_c^a K f(x)dx = K \int_c^a f(x)dx$

C)  $\int_c^a f(x)dx = \int_b^a f(x)dx + \int_c^b f(x)dx$

D)  $\int_{-a}^a f(x)dx = 2 \times \int_0^a f(x)dx$

- iii) If  $y = f(x)$  is defined at  $x = a$ . Which one of the following gives the necessary condition(s) for the curve  $y = f(x)$  to be continuous at  $x = a$ .
- A)  $f'(a) = 0$  B)  $f(a) = f'(a)$   
C)  $\lim_{x \rightarrow a} f(x) = f(a) = \lim_{x \rightarrow a} f'(x)$  D)  $\lim_{x \rightarrow a} f(x) = f'(a)$
- iv) A parabola has parametric equations  $x = 2p$ ,  $y = -2 - 2p^2$ . This parabola has :
- A) Vertex  $(0, -2)$  and Focus  $(0, -2\frac{1}{2})$  B) Vertex  $(0, 2)$  and Focus  $(0, -1\frac{1}{2})$   
C) Vertex  $(0, -2)$  and Focus  $(0, -1\frac{1}{2})$  D) Vertex  $(0, 2)$  and Focus  $(0, -2\frac{1}{2})$
- v) The hyperbola  $y = \frac{2x-3}{x+6}$  has a :
- A) vertical asymptote with equation  $x = -6$  and a horizontal asymptote with equation  $y = 1\frac{1}{2}$   
B) vertical asymptote with equation  $x = -6$  and a horizontal asymptote with equation  $y = 2$   
C) vertical asymptote with equation  $x = -6$  and a horizontal asymptote with equation  $y = 0$   
D) vertical asymptote with equation  $x = -6$  and a horizontal asymptote with equation  $y = -\frac{1}{2}$
- vi) A student uses the substitution  $y = \sqrt{x}$  to find  $\int \frac{dx}{\sqrt{x(1-x)}}$ .

Which one of the following represents a equivalent definite integral.

- A)  $2 \int \frac{dy}{\sqrt{1-y^2}}$       B)  $\frac{1}{2} \int \frac{dy}{\sqrt{1-y^2}}$
- C)  $2 \int \frac{y dy}{\sqrt{1-y^2}}$       D)  $\frac{1}{2} \int \frac{y dy}{\sqrt{1-y^2}}$

**QUESTION 2****9 marks***(start a new page)*

- a) The point  $P(-1, -3)$  divides the interval joining  $A(5, 9)$  and  $B(-3, -7)$  internally in the ratio  $k : 1$ . Find  $k$  (3)
- b) Solve  $\frac{4x+3}{2x-1} < 1$  (3)
- c) Find the acute angle between the two curves  $y = x^3$  and  $y = (2x-1)^2$  at their point of intersection  $(1, 1)$ . Give your answer correct to the nearest degree. (3)

**QUESTION 3****9 marks***(start a new page)*

- a) The equation  $2x^3 - 9x^2 + 12x - 4 \cdot 01 = 0$  has a root near  $x = 2$
- (i) If  $x = 2$  is chosen as the initial approximation, explain why Newton's Method for finding an improved approximation fails. (2)
- (ii) Use Newton's Method once and  $x = 1 \cdot 5$  as an initial approximation to the root of  $2x^3 - 9x^2 + 12x - 4 \cdot 01 = 0$  to find an improved approximation to the root. (write your answer correct to 2 d.p.) (2)
- b)
- (i) Show that the quadratic polynomial  $3x^2 + 2x + 1$  is positive definite. (1)
- (ii) If  $P(x) = x^3 + x^2 + x - 8$ , use the sign derivative of  $P(x)$  to explain why the equation  $P(x) = 0$  has only one root for all real values of  $x$ . (1)
- (iii) Given that this root lies between  $x = 1 \cdot 5$  and  $x = 2$ , use the Halving the Interval Method for approximating a root once to show that the root lies more accurately between  $x = 1 \cdot 5$  and  $x = 1 \cdot 75$  (2)
- (iv) Is the root closer to  $x = 1 \cdot 5$  or  $x = 1 \cdot 75$ . Give reasons for your answer. (1)

**QUESTION 4****9 marks***(start a new page)*

- a) Show, by process of mathematical induction, that  $3^{4n} - 1$  is a multiple of 80 for all positive integral values of  $n$  (3)

- b) Use mathematical induction to prove that

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1) \times 2^n \text{ for all positive integral values of } n \quad (4)$$

- c) A student is asked to use Mathematical Induction to prove that  $5^n > 3^n + 4^n$  for integers  $n \geq 3$ .

His incomplete solution to the proof is given below.

**Complete Step 3 of the proof on your own paper** (2)

Step 1 Prove true for  $n = 1$

|                      |                      |
|----------------------|----------------------|
| L.H.S. = $5^n$       | R.H.S. = $3^n + 4^n$ |
| = $5^3$ when $n = 3$ | = $3^3 + 4^3$        |
| = 125                | = 91                 |

$$\therefore \text{L.H.S.} > \text{R.H.S. for } n = 3$$

$$\therefore 5^n > 3^n + 4^n \text{ for } n = 3$$

Step 2 Assume true for  $n = k$

i.e. assume  $5^k > 3^k + 4^k$

Step 3 Prove true for  $n = k + 1$ , if true for  $n = k$

i.e. prove  $5^{k+1} > 3^{k+1} + 4^{k+1}$

.....

.....

.....

.....

Step 4 Since true for  $n = 1$  .....

## QUESTION 5

9 marks

(start a new page)

- a) The normal at the point  $P(4t, 2t^2)$  on the parabola  $x^2 = 8y$  intersects the  $y$  axis at  $Q$ .
- (i) Show that the gradient of the normal at  $P$  is  $-\frac{1}{t}$  (1)
- (ii) Show that the equation of the normal at  $P$  is  $x + ty = 4t + 2t^3$  (1)
- (iii) Find, in terms of  $t$ , the coordinates of  $Q$ . (1)
- (iv) State the **parametric equations** of the locus of the midpoint of  $PQ$ . (1)
- b) The points  $P(2p, p^2)$  and  $Q(2q, q^2)$  lie on the parabola  $x^2 = 4y$ .  
The chord  $PQ$ , when produced, always passes through the point  $A(2, 0)$ .
- (i) Noting that  $p \neq q$ , show that  $p + q = pq$ . (2)
- (ii) Find the co-ordinates of  $M$ , the midpoint of  $PQ$  (1)
- (iii) Find the **cartesian equation** of the locus of  $M$  (2)

## QUESTION 6

9 marks

(start a new page)

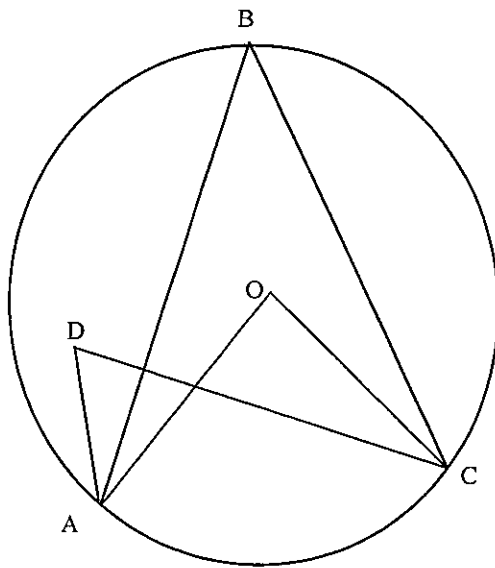
- a) Evaluate the definite integral  $\int \frac{t \, dt}{\sqrt{4-t}}$  using the substitution  $t = 4 - w^2$  (3)
- b) Find  $\int \frac{x \, dx}{\sqrt{1-9x^2}}$  using the substitution  $u = \sqrt{1-9x^2}$  (3)
- c) Use the substitution  $u = x+1$  to evaluate  $\int_0^3 \frac{x-2}{\sqrt{x+1}} \, dx$  (3)

QUESTION 7

9 marks

(start a new page)

a)

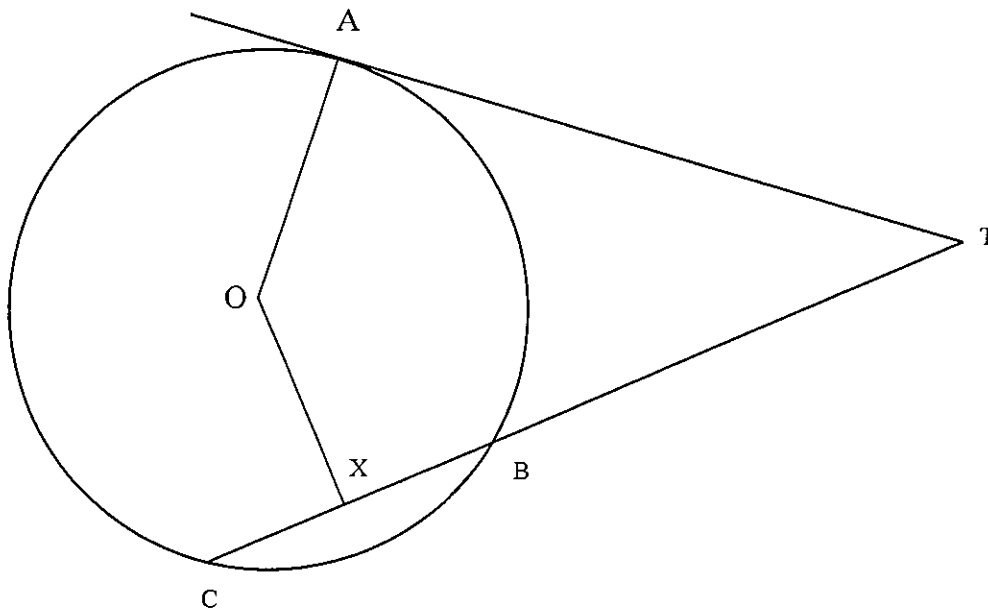


Points  $A$ ,  $B$  and  $C$  lie on the circumference of a circle with centre  $O$ .  
The point  $D$  lies inside the circle.  $\angle ABC = 17^\circ$  and  $\angle ADC = 34^\circ$

(i) Find  $\angle AOC$  giving reasons. (1)

(ii) State why  $ADOC$  is a cyclic quadrilateral. (1)

b)

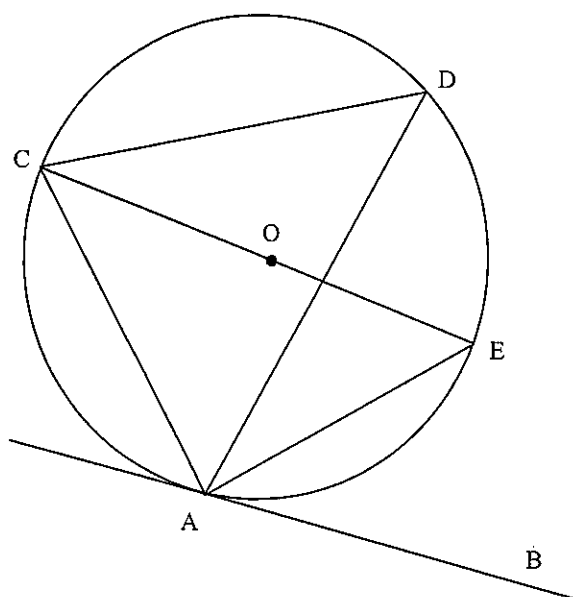


$A$ ,  $B$  and  $C$  are three points on a circle with centre  $O$ .

The tangent at  $A$  meets the chord  $CB$  produced at  $T$ .  $X$  is the midpoint of  $CB$ .

Prove that  $AOXT$  is a cyclic quadrilateral. (3)

c)



$AB$  is a tangent and  $CE$  is a diameter to a circle with centre  $O$ .

$\angle BAE$  is  $48^\circ$  and  $D$  lies on the circumference as show in the diagram.

- (i) Find the size of  $\angle ACE$ , giving reasons. (1)
- (ii) Find the size of  $\angle ADC$ , justifying your answer with reasoning. (3)

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x$ ,  $x > 0$

SOLUTIONSQuestion 1

(i) C (ii) D (iii) C

(iv) A (v) B (vi) A

Question 2 a) A(5, 9) B(-3, -7)

$$k : 1$$

$$\therefore \frac{-3k+5}{k+1} = -1$$

$$-3k+5 = -k-1$$

$$6 = 2k$$

$$\therefore k = 3$$

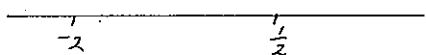
b) Critical values occur when

$$2x-1=0 \\ x=\frac{1}{2}$$

and when

$$\frac{4x+3}{2x-1} = 1$$

$$4x+3 = 2x-1 \\ 2x = -4 \\ x = -2$$

FALSE When  $x < -2$  (say  $x = -3$ )TRUE When  $-2 < x < \frac{1}{2}$  (say  $x = 0$ )FALSE When  $x > \frac{1}{2}$  (say  $x = 1$ ) $\therefore$  Solution is  $-2 < x < \frac{1}{2}$ 

c)

$$y = x^3$$

$$\frac{dy}{dx} = 3x^2$$

$$m_1 = 3 \text{ when } x=1$$

$$y = (2x-1)^2$$

$$\frac{dy}{dx} = 2(2x-1) \cdot 2$$

$$m_2 = 4$$

$$\text{when } x=1$$

Let  $\theta$  be the required acute  $\angle$ .

$$\therefore \tan \theta = \left| \frac{m_2 - m_1}{1 + m_2 m_1} \right|$$

$$\tan \theta = \left| \frac{4-3}{1+(4)(3)} \right|$$

$$\tan \theta = \frac{1}{13}$$

$$\theta = 4^\circ \text{ (to the nearest degree)}$$

Question 3(a) (i) Let  $P(x) = 2x^3 - 9x^2 + 12x - 4.01$ 

$$P'(x) = 6x^2 - 18x + 12$$

$$P'(2) = 24 - 36 + 12 \\ = 0$$

 $\therefore$  Stationary Pt. exists at  $x=2$ 

$\therefore$  Tangent at  $x=2$  does not cross the  $x$ -axis.  
The  $x$ -intercept of the tangent is the improved approximation.  
Thus an improved approximation cannot be found.

(ii) If  $x_1 = 1.5$  is the initial approx.

$$P(x_1) = P(1.5) \\ = 0.49$$

$$P'(x_1) = P'(1.5) \\ = -1.5$$

If  $x_2$  is improved approx. then 
$$x_2 = x_1 - \frac{P(x_1)}{P'(x_1)} \\ = 1.5 + \frac{0.49}{1.5} \\ = 1.83 \text{ (to 2d.p.)}$$

$$\begin{aligned} b) \quad (i) \quad \Delta &= b^2 - 4ac \\ &= 4 - 4(3)(1) \\ &= -8 < 0 \end{aligned}$$

Since  $a = 3 > 0$  and  $\Delta < 0$  expression is +ve definite

$$(ii) \quad P'(x) = 3x^2 + 2x + 1 > 0$$

$\therefore$  Curve is always increasing  
Since  $P(x)$  is continuous for all  $x$   
the equation  $P(x) = 0$  has only one root

$$(iii) \quad P(1.5) = -0.875 < 0 \quad P(2) = 6$$

Let  $x_0 = \frac{1.5+2}{2}$  be new approx.

$$x_0 = 1.75$$

$$P(1.75) = 2.171875 > 0$$

$P(1.75)$  and  $P(1.5)$  are opposite in sign  
 $\therefore$  Root lies between  $x = 1.5$  and  $x = 1.75$

$$\begin{aligned} (iv) \quad \text{Consider } P\left(\frac{1.5+1.75}{2}\right) &= P(1.625) \\ &= 0.55 > 0 \end{aligned}$$

$\therefore$  Root lies between  $x = 1.625$  and  $x = 1.5$   
 $\therefore$  Root is closer  $x = 1.5$  than  $x = 1.75$

#### Question 4

a) Prove true for  $n=1$

$$\begin{aligned} 3^{4n} - 1 &= 3^4 - 1 \quad \text{when } n=1 \\ &= 80 \end{aligned}$$

Which is a multiple of 80  
 $\therefore$  true for  $n=1$

Assume true for  $n=k$

$$\begin{aligned} \text{i.e. } 3^{4k} - 1 &= 80M \quad (\text{where } M \text{ is a +ve integer}) \\ 3^{4k} &= 80M + 1 \end{aligned}$$

Prove true for  $n=k+1$ , if true for  $n=k$

$$\begin{aligned} \text{i.e. Prove } 3^{4(k+1)} - 1 &\text{ is a multiple of 80} \\ 3^{4(k+1)} - 1 &= 3^{4k+4} - 1 \\ &= 3^4 \cdot 3^{4k} - 1 \\ &= 81(80M+1) - 1 \quad \text{using assumption} \\ &= 81 \times 80M + 81 - 1 \\ &= 80(81M+1) \end{aligned}$$

Which is a multiple of 80 since  
 $(81M+1)$  is a +ve integer

$\therefore$  true for  $n=k+1$ , if true for  $n=k$

Since true for  $n=1$   $\therefore$  true for  $n=1+1=2$

Since true for  $n=2$   $\therefore$  true for  $n=2+1=3$

and so on.

$\therefore$  true for all given 'n'

Question 4 (b) Prove true for  $n=1$

$$L.H.S. = n \times 2^{n-1}$$

$$= 1 \times 2^0 \text{ when } n=1$$

$$= 1$$

$\therefore$  true for  $n=1$

$$R.H.S. = 1 + (n-1) \times 2^n$$

$$= 1 + 0 \text{ when } n=1$$

$$= 1$$

$$= L.H.S.$$

Assume true for  $n=k$

$$\text{i.e. } 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1) \times 2^k$$

Prove true for  $n=k+1$ , if true for  $n=k$

$$\text{i.e. Prove } 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} + (k+1) \times 2^k = 1 + k \times 2^{k+1}$$

$$L.H.S. = 1 + (k-1) \times 2^k + (k+1) \times 2^k \text{ using assumption}$$

$$= 1 + 2^k (k-1+k+1)$$

$$= 1 + 2^k \times 2k$$

$$= 1 + k \times 2^{k+1}$$

$$= R.H.S.$$

$\therefore$  true for  $n=k+1$  if true for  $n=k$

Since true for  $n=1$   $\therefore$  true for  $n=1+1=2$

Since true for  $n=2$   $\therefore$  true for  $n=2+1=3$

and so on

$\therefore$  true for all give 'n'

Question 4 c)

$$5^{k+1} = 5 \times 5^k$$

$$\therefore 5^{k+1} > 5(3^k + 4^k) \text{ using assumption}$$

$$5^{k+1} > 5 \times 3^k + 5 \times 4^k$$

$$5^{k+1} > 3 \times 3^k + 4 \times 4^k \text{ since } 3 < 5 \text{ and } 4 < 5$$

$$5^{k+1} > 3^{k+1} + 4^{k+1}$$

$\therefore$  true for  $n=k+1$  if true for  $n=k$

Question 5

$$a) (i) y = \frac{1}{8} x^2$$

$$\frac{dy}{dx} = \frac{1}{4} x$$

$$= \frac{1}{4} \times 4t \text{ at } P$$

$$= t$$

(gradient of tangent at P)

$\therefore$  gradient of normal at P is  $-\frac{1}{t}$

$$(ii) \text{ Equation of normal is } y - 2t^2 = -\frac{1}{t}(x - 4t)$$

$$\text{i.e. } ty - 2t^3 = -x + 4t$$

$$x + ty = 4t + 2t^3$$

$$(iii) \text{ At } y \text{ axis } x = 0$$

$$\therefore ty = 4t + 2t^3$$

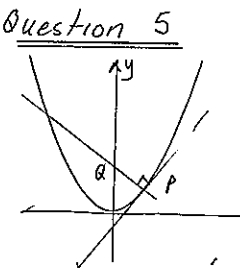
$$y = 4 + 2t^2$$

$$\therefore Q(0, 4 + 2t^2)$$

$$(iv) \text{ Midpoint of } PQ \text{ is } \left( \frac{4t}{2}, \frac{2t^2 + 4 + 2t^2}{2} \right)$$

$$= (2t, 2t^2 + 2)$$

$\therefore$  Parametric equations are  $x = 2t$ ,  $y = 2t^2 + 2$



$$b) \quad (i) \quad m_{AP} = m_{AQ}$$

$$\frac{p^2}{2p-2} = \frac{q^2}{2q-2}$$

$$2p^2q - 2p^2 = 2q^2p - 2q^2$$

$$2p^2q - 2q^2p = 2p^2 - 2q^2$$

$$2pq(p-q) = 2(p-q)(p+q)$$

$$pq = p+q \quad \text{since } p \neq q$$

$$(ii) \quad M \text{ is } \left( \frac{2p+2q}{2}, \frac{p^2+q^2}{2} \right)$$

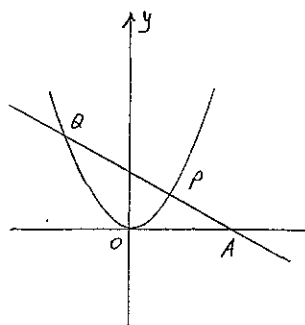
$$= \left( p+q, \frac{p^2+q^2}{2} \right)$$

(iii) Parametric equations are

$$x = p+q \quad \text{and} \quad y = (p+q)^2 - 2pq$$

$$\therefore y = x^2 - 2(p+q) \quad \text{using (i)}$$

$$y = x^2 - 2x \quad \text{is the cartesian equation}$$



Question 6

$$a) \quad t = 4 - w^2 \rightarrow w^2 = 4 - t$$

$$\frac{dt}{dw} = -2w$$

$$\therefore \int \frac{t dt}{\sqrt{4-t}} = \int \frac{4-w^2}{\sqrt{w^2}} \times -2w dw$$

$$= 2 \int w^2 - 4 dw$$

$$= 2 \left[ \frac{w^3}{3} - 4w \right] + C$$

$$= \frac{2}{3} \sqrt{(4-t)^3} - 8\sqrt{4-t} + C$$

$$b) \quad u = (1-9x^2)^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2} (1-9x^2)^{-\frac{1}{2}} \times -18x$$

$$= \frac{-9x}{\sqrt{1-9x^2}}$$

$$\frac{dx}{du} = \frac{u}{-9x}$$

$$\therefore \int \frac{x dx}{\sqrt{1-9x^2}} = \int \frac{x}{u} \times \frac{u}{-9x} du$$

$$= -\frac{1}{9} \int 1 du$$

$$= -\frac{1}{9} u + C$$

$$= -\frac{1}{9} \sqrt{1-9x^2} + C$$

Question 6

c)  $u = x+1$  when  $x=3$ ,  $u=4$

$$\frac{du}{dx} = 1 \rightarrow \frac{dx}{du} = 1 \quad x=0, u=1$$

$$\begin{aligned}
 \therefore \int_0^3 \frac{x-2}{\sqrt{x+1}} dx &= \int_1^4 \frac{u-3}{\sqrt{u}} \times 1 du \\
 &= \int_1^4 \frac{u}{u^{\frac{1}{2}}} - \frac{3}{u^{\frac{1}{2}}} du \\
 &= \int_1^4 u^{\frac{1}{2}} - 3u^{-\frac{1}{2}} \\
 &= \left[ \frac{2u^{\frac{3}{2}}}{3} - 6u^{\frac{1}{2}} \right]_1^4 \\
 &= \left[ \frac{2}{3} \times 8 - 6 \times 2 - \left( \frac{2}{3} - 6 \right) \right] \\
 &= \frac{14}{3} - 6 \\
 &= -\frac{4}{3}
 \end{aligned}$$

Question 7

a)

(i)  $\angle AOC = 2 \times \angle ABC$  ( $\angle$  at the centre is twice the angle at the circumference, standing on the same arc)  
 $= 2 \times 17^\circ$   
 $= 34^\circ$

(ii)  $\angle AOC = \angle ADC$   
 $= 34^\circ$

Since the interval AC subtends two equal  $\angle$ 's at O and D then the end points of the interval, A & C and the vertices of the angles, O & D are concyclic i.e. ADOC is a cyclic quadrilateral

Question 7

b)  $\angle AOT = 90^\circ$  (tangent is perpendicular to the radius drawn at the point of contact)

$\angle AXB = 90^\circ$  (interval drawn from the centre of a circle to the midpoint of a chord is perpendicular to the chord)

$\therefore \angle AOT + \angle AXB = 180^\circ$

Since the opposite angles of the quadrilateral AOTX are supplement, AOTX must be a cyclic quadrilateral.

c) (i)  $\angle ACE = \angle ABE$  ( $\angle$  between a tangent and a chord drawn at the point of contact is equal to the angle in the alternate segment)  
 $= 48^\circ$

(ii)  $\angle CAE = 90^\circ$  (angle in a semi-circle is a right angle)

$\angle CEA = 180 - (90 + 48)^\circ$  ( $\angle$  sum of  $\triangle ACE$ )  
 $= 42^\circ$

$\angle ADC = \angle CEA$  (angles at the circumference standing on the same arc are equal)